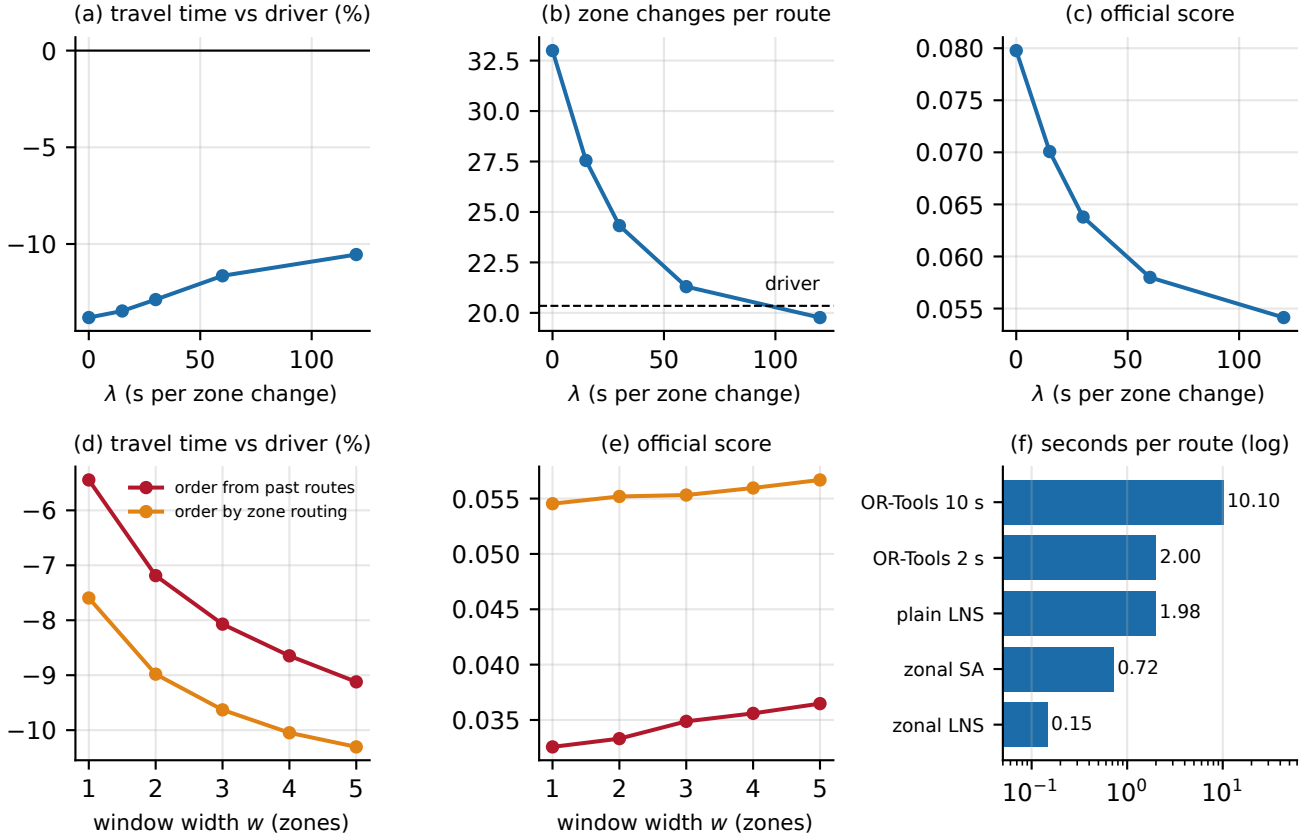
(a) travel time vs driver (%)
0
−5
−10
0
50
100
λ (s per zone change)
(b) zone changes per route
32.5
30.0
27.5
25.0
22.5
20.0
driver
0
50
100
λ (s per zone change)
(c) official score
0.080
0.075
0.070
0.065
0.060
0.055
0
50
100
λ (s per zone change)
(d) travel time vs driver (%)
order from past routes
order by zone routing
−6
−7
−8
−9
−10
1
2
3
4
5
window width w (zones)
(e) official score
0.055
0.050
0.045
0.040
0.035
1
2
3
4
5
window width w (zones)
(f) seconds per route (log)
OR-Tools 10 s
OR-Tools 2 s
plain LNS
zonal SA
zonal LNS
10.10
2.00
1.98
0.72
0.15
10^−1
10^0
10^1